

Hussein Baydoun

Bint Jbeil, Lebanon
husseinbaydoun1711@icloud.com
Phone Number: +961 71 524 984
GitHub: HusseinBYDN

EDUCATION

Phoenicia University (PU) - Sarafand, Lebanon

October 2022 - Present

Bachelor of Science in Computer Science

Key Courses: Algorithms, Java, Web Development, Software Engineering, C++

Expected Graduation Date: February 2026

Al Mahdi Schools – Bint Jbeil, Lebanon

Awarded in June 2022

Lebanese Baccalaureate in General Sciences

Internships

Makhzoumi Foundation – IT Program – Beirut, Lebanon

September 2025

Software Development Intern

- Designed and developed SkillBridge, a full-stack internship management web application.
- Built and deployed back-end services using ASP.NET Core 9.0 and Entity Framework Core, including database schema design, migrations, and RESTful API endpoints.
- Developed responsive front-end interfaces using Razor Pages and integrated them with APIs via AJAX and jQuery.
- Implemented Role-Based Access Control (RBAC) to differentiate Admin, Coordinator, and Student workflows.
- Enhanced system security using HTTPS, bcrypt password hashing, validation, and OWASP best practices.
- Collaborated in an Agile environment with weekly sprints, code reviews, GitHub version control, GitFlow branching, testing, and Azure deployment preparation.

Workshops & Projects

AI-Based Early Myocardial Infarction Detection (FYP) – PU

March 2026 - Present

- Developed an applied AI research project for early myocardial infarction detection using 12-lead ECG signals from the PTB-XL dataset.
- Built and evaluated CNN baselines, fixed reduced-observation strategies, and adaptive reinforcement-learning acquisition under shared lead-second budgets.
- Implemented partial observation masking for lead/time windows and evaluated results using AUROC, diagnostic metrics, calibration outputs, and budget-performance comparisons.

Topological Data Analysis Extension for ECG/MI Detection – PU

2026 - Present

- Added a complementary TDA branch using delay embedding, point-cloud generation, persistent homology, persistence diagrams, and classical ML classifiers.
- Completed the Task A implementation stage and prepared the same lead-second budget framework for Tasks B and C.

Credit Card Fraud Detection Using Neural Network – PU

January 2026

- Built a TensorFlow/Keras neural-network classifier to detect fraudulent credit card transactions in a highly imbalanced dataset.
- Handled class imbalance using preprocessing, class weighting, and stratified train/validation/test workflows.
- Evaluated performance using precision, recall, F1-score, ROC-AUC, PR-AUC, and confusion matrix.

Linux Character Device Driver Project – PU

January 2026

- Implemented a Linux kernel module for device communication.
- Developed read/write operations and a testing program for user-kernel interaction.
- Applied mutex synchronization techniques to support safe access.

nutrition_byzahraa Full-Stack Dietitian Platform

2026

- Designed and developed a full-stack dietitian services platform with public pages, user workflows, and admin management screens.
- Implemented consultation requests, plan requests, feedback forms, blog content, contact messages, and user status tracking.
- Designed the MySQL database structure for users, admins, requests, feedback, services, blog posts, and contact messages.

Flower Shop Database System – PU

May 2025

- Designed and implemented a complete relational database system for a flower shop using MySQL.
- Built tables, relationships, SQL queries, EER diagrams, schemas, and project documentation.
- Demonstrated proficiency in database normalization and relational schema design.

Connecting Front-End to Back-End – Codecademy

April 2025

- Completed an intermediate course focused on integrating front-end and back-end technologies, RESTful APIs, MVC architecture, React, and Node.js.

Software Engineering Project – PU

February 2024

- Designed and developed a full software system and completed full project documentation.

Python Workshop – PU

August 2023

- Completed training in Python programming, including data structures, object-oriented programming, and file handling.

Skills and Interests

Linguistic Skills: Fluent in English and Arabic

AI / Machine Learning: Python, TensorFlow/Keras, PyTorch-based CNN pipelines, neural networks, reinforcement learning, class weighting, model evaluation, train/validation/test workflows

Evaluation & Data: AUROC, precision, recall, specificity, F1-score, confusion matrices, calibration curves, budget-performance comparison, 12-lead ECG preprocessing, PTB-XL metadata handling, lead/time budgets

Topological Data Analysis: delay embedding, point clouds, persistent homology, persistence diagrams, H0/H1 features, topology-based ECG representation, classical ML classifiers

Web & Backend Development: HTML, CSS, JavaScript, React, ASP.NET Core 9.0, Razor Pages, RESTful APIs, AJAX, jQuery, Node.js exposure

Database Systems: MySQL, SQL Server, Entity Framework Core, LINQ, schema design, migrations, relationships, normalization, EER diagrams, SQL queries

Security, Tools & Methodologies: RBAC, bcrypt password hashing, HTTPS, OWASP best practices, Postman, GitHub (GitFlow), Agile/Scrum, Azure App Service preparation

Systems & Mobile: C/C++, Java, Dart, Flutter, Linux kernel module basics, device driver read/write operations, mutex synchronization

Other: MS Office (Word, Excel, PowerPoint)